

UEC644: ANALOG IC DESIGN				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: The goal is to achieve a basic understanding and knowledge of the driving and limiting factors in circuit performance, of circuit design techniques, and of technology issues important to integrated amplifier circuits. To familiarize the design and analysis of basic analog integrated circuits i.e. single ended amplifiers, differential amplifiers, current sources and mirrors, reference circuits, etc. in a standard flow with consideration of performance and power. The course will also familiarize with the issues like noise analysis, OP-Amp design, stability and compensation				
Syllabus Basic MOS Device Physics: MOS IV Characteristics, Second order effects, Short-Channel Effects, MOS Device Models, Review of Small Signal MOS Transistor Models. Single Stage Amplifiers: Common Source Stage, Source Follower, Common Gate Stage, Cascode, Folded Cascode. Differential Amplifier: Single ended and Differential Operation, Qualitative and Quantitative Analysis of Differential pair, Common Mode response. Current Sources and Mirrors: Current Sources, Basic Current Mirrors, Cascode Current Mirrors, Large Signal and Small-Signal analysis. Frequency Response of Amplifiers: Miller Effect, Association of Poles with nodes, Frequency Response of all single stage amplifiers. Voltage References: Different Configurations of Voltage References, Major Issues, Supply Independent Biasing, Temperature-Independent References. Feedback: General Considerations, Topologies, Effect of Loading. Operational Amplifier: General Considerations, Theory and Design, Performance Parameters, Single-Stage Op Amps, Two-Stage Op Amps, Design of 2-stage MOS Operational Amplifier, Gain Boosting, Comparison of various topologies, slew rate, Offset effects, PSRR. Stability and Frequency Compensation: General Considerations, Multi-pole systems, Phase Margin, Frequency Compensation, Compensation Techniques. Familiarization with standard: <i>IEEE 802.15.4.</i>				
Laboratory Work: Review of Mentor Tools; Analysis of Various Analog Building Blocks such as, Current Sources, Current Mirrors, Differential Amplifier, Output Stages; Design and Analysis of Op-Amp (Closed loop and open loop) and its Characterization, Switched-Capacitor Integrator.				
Course Learning Objectives (CLO) The students will be able to: 1. Utilize the MOS transistor in analog circuits.				

2. Analyze and design the single-stage amplifiers.
3. Analyze and design current sources/sinks/mirrors.
4. Analyze the characteristics of OP-AMP.

Text Books

1. Razavi, B., *Design of Analog CMOS Integrated Circuits*, Tata McGraw Hill (2008).
2. Gregorian, R. and Temes, G.C., *Analog MOS Integrated Circuits for Signal Processing*, John Wiley (2004).

Reference Books

1. Allen, P.E. and Holberg, D.R., *CMOS Analog Circuit Design*, Oxford University Press (2002) 2nd ed.
2. Johns, D.A. and Martin, K., *Analog Integrated Circuit Design*, John Wiley(2008).
3. Gray, P.R., Hurst, P.J., Lewis, S.H., and Meyer, R.G., *Analysis and Design of Analog Integrated Circuits*, John Wiley (2001) 5th ed.